Jaxon Poentis

808-365-7008 | j.k.poentis@wustl.edu | linkedin.com/in/jaxon-poentis/ | github.com/jaximus808 | jaxonp.com

Education

Washington University in St. Louis — St. Louis, MO
Bachelor of Science in Computer Science + Math — *May 2027*

- **Awards:** Eagle Scout, Anheuser-Busch Scholar, Dean's List
- **Coursework:** Parallel Programming, Cloud Computing, System Software, OOP, Rapid Prototyping

Experience

Capital One — June 2026 – Present
Software Engineering Intern — *McLean, VA*

- Built a self-serve FastAPI web app with a team of 5, enabling enterprise teams to search across security groups by CIDR and rule, replacing slow manual lookups across the cloud network environment.
- Implemented PostgreSQL query optimization using GiST indexes to search 1.5M+ security groups and rules in under a second, delivering near-instant self-serve results at scale.

Tesla — Jan. 2026 – May 2026
Software Engineering Intern — *Fremont, CA*

- Created, led, and deployed an AI agent platform (Python, FastAPI, LangGraph, MongoDB, WebSockets) across energy services, adopted by billing, support, and application teams, building 12+ tool interfaces for automated data retrieval and business workflow execution on the residential energy platform.
- Led a cross-team effort to design and ship Tesla account-based authentication for the agent platform, integrating Cerberus auth with account passthrough to enable secure, scoped tool execution.
- Engineered asynchronous workflows across distributed .NET microservices to orchestrate energy lease transfers and propagate associated account data to production, serving Tesla's entire U.S. energy customer base.
- Deployed and managed 10+ services across Kubernetes pods with Jenkins CI/CD pipelines, handling container lifecycle, scaling, and release coordination across staging and production environments.

Spectrum — May 2025 – Aug. 2025
Software Engineering Intern — *St. Louis, MO*

- Built a Python coupling analysis tool with custom search algorithms and AST parsing to map nested Java class dependencies, reducing refactoring errors and saving 2–4 days of manual validation per development cycle.
- Developed backend controller logic and fulfillment workflow integrations to trigger automated email notifications on status changes, enabling faster cross-team collaboration and response times.

Projects

Tandem – Canvas for Humans & Agents | *Go, TypeScript, WebSockets, MCP* — May 2026 – Present

- Built and shipped a real-time collaborative canvas (tandemcanvas.com) where humans and AI agents co-edit shared maps, docs, sheets, and roadmaps; engineered a Go WebSocket hub for low-latency multi-client sync and a Node MCP server exposing the canvas as agent-callable tools, backed by Postgres and deployed on GCP via Docker and GitHub Actions.

ANDR – Robotics Agent Framework | *C++, Python, ROS 2, LangGraph* — Feb. 2026 – Present

- Built an open-source ROS 2 framework where an LLM controls a robot through strictly decoupled layers; wrote a C++ tool-manager that dynamically discovers and dispatches skill calls to independent action servers, a C++ behavior-tree brain, and a Python ReAct agent, with Gazebo simulation and a FastAPI/WebSocket UI.

Leadership

Google Developer Group – WashU — Aug. 2025 – Present
Core Team Lead — *St. Louis, MO*

- Led a team of 8 students through a semester-long project, mentoring on system design and engineering practices while organizing workshops and tech events for the broader WashU CS community.

Technical Skills

Languages: Java, Python, C++, C, C#, Go, SQL (Postgres), JavaScript, TypeScript, HTML/CSS
Frameworks: Node.js, .NET, React.js, Phaser.js, Express.js, Gin, FastAPI, ROS2, OpenCV, Unity
Technologies: Docker, Kubernetes, Jenkins, Supabase, Firebase, Vultr, MongoDB, PostgreSQL, UDP/TCP, Git, Linux, OAuth